

Dummy Variable: A qualitative variable which takes values that can be divided into categories.

Time Series Data: This refers to data measured over a period of time at regular intervals.

14.10 SOME USEFUL BOOKS

Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill

Maddala, G. S. and K. Lahiri (2012), *Introduction to Econometrics*, Fourth Edition, Wiley

Verbeek, M. (2012), “*A Guide to Modern Econometrics*”, Fourth Edition, John Wiley & Sons

Wooldridge, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

14.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Dummy variable is suggested where explanatory variable is qualitative in nature. Go through Section 14.3.
- 2) It will lead to a situation of perfect multicollinearity. Go through Unit 10 to find the consequences.

Check Your Progress 2

- 1) The procedure of Chow test is given in Section 14.2. On that basis test the model.
- 2) Go through Section 14.4 and formulate the model with differential intercept and slope coefficients. Test the model based on t-test of the dummy variable coefficients.

UNIT 15 QUALITATIVE DEPENDENT VARIABLES IN OLS MODELS*

Structure

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15.0 OBJECTIVES

After going through this Unit, you will be in a position to

- explain how qualitative (dummy) variables can be incorporated as dependent variable in a regression model;
- describe the advantages and disadvantages of linear probability model (LPM);

*Dr. Akshaya Kumar Mohanty, Associate Professor, Mizoram University, Aizawl

- explain how logit and probit models can remove the disadvantages of LPM;
- calculate and interpret marginal effects in LPM, Logit model, and Probit model; and
- interpret Stata software generated output for all three models.

15.1 INTRODUCTION

In Unit 14 we discussed the use of qualitative variables (dummy variables) as explanatory variables in a regression model. Such qualitative variables can assume binary values of 0 and 1. A question may be coming to your mind: Can we use a dummy variable as explained variable (or, dependent variable) in a regression model? The answer is: Yes.

In this Unit we will look at regression models where the dependent variable, Y , is a qualitative variable. Such models are called ‘limited dependent variable models’, or dummy dependent variable models. The term ‘limited dependent variable’ was first used by James Tobin in 1958 to denote regression models where the dependent variable is constrained to be non-negative. As you know, nominal variables can assume more than two categories. If the dummy dependent variable has exactly two categories, it is called a binary (or dichotomous) variable. Otherwise, it is called *polychotomous variable* (containing more than two categories). The present Unit is introductory in nature, and we will focus on binary cases where a dummy variable can take only two values, 0 and 1.

If the *dependent* variable of the model is a dummy variable, the usual OLS method will not be applicable. Instead, the method of *maximum likelihood estimation* is used. This is because when the dependent variable is a dummy variable, the objective is to find out the *maximum probability* (hence the name *maximum likelihood*) of occurrence of the given set of data.

Binary choice models are commonly used for micro analysis in social sciences and medical research. There are three types of ‘limited dependent variable models’. These are: (i) Linear Probability Models (LPM), (ii) Logit Models, and (iii) Probit Models. The last two models have quite similar probability density functions, similar estimation procedures, and they are often used interchangeably.

15.2 LINEAR PROBABILITY MODEL

A linear probability model (LPM) is a regression model where the dependent variable (regressand) is a binary (or, dummy) variable, and the independent variables are either dummy or continuous.

15.2.1 Specification of the Model

To fix ideas, let us consider the following linear probability regression model on determinants of house ownership

$$Y_i = \alpha_0 + \alpha_1 X_i + \alpha_2 D_i + u_i \quad \dots (15.1)$$

where Y_i = 1, if a person owns a house
 = 0, if he/she does not own a house
 X_i = is family income
 D_i = 1, if the person is male, and 0 otherwise.

Equation (15.1) looks like a linear regression model but as the dependent variable (Y_i) is binary, or dichotomous, it is called a linear probability model (LPM). The conditional expectation of Y_i given by X_i and D_i , $E(Y_i|X_i, D_i)$, can be interpreted as the *conditional probability* that the event will occur given X_i and D_i , that is, $\Pr(Y_i = 1|X_i, D_i)$. Thus, in our example, $E(Y_i|X_i, D_i)$ gives the probability of a person owning a house and whose income is given by X_i and whose gender is identified by D_i . Equation (15.1) can be estimated by using the OLS method. The OLS method, however, has certain drawbacks.

The justification of the name LPM for model (15.1) can be seen as follows: By assuming $E(u_i) = 0$ and taking expected value of equation (15.1), we obtain,

$$E(Y_i|X_i, D_i = 1) = \alpha_0 + \alpha_1 X_i + \alpha_2 \quad \dots (15.2)$$

Suppose the probability that the event will occur, that is, $\Pr(Y_i = 1) = p_i$. Then, the probability that the event does not occur $\Pr(Y_i = 0) = (1 - p_i)$.

Taking the expected value

$$\begin{aligned} E(Y_i|X_i, D_i) &= E(\alpha_0 + \alpha_1 X_i + \alpha_2) \\ &= 1(p_i) + 0(1 - p_i) \end{aligned} \quad \dots (15.3)$$

Since the probability p_i must remain between 0 and 1, we have the restriction on the conditional expectation that

$$0 \leq E(Y_i|X_i, D_i) \leq 1 \quad \dots (15.4)$$

The conditional expectation of the model (15.1) can, in fact, be interpreted as the conditional probability of Y_i . In general, the expectation of a Bernoulli random variable is the probability that the random variable equals 1. If there are n independent trials, each with a probability p of success and probability $(1 - p)$ of failure, and X of these trials represent the number of successes, then X is said to follow the binomial distribution. The mean of the binomial distribution is np and its variance is $np(1-p)$. Thus, the OLS method can be easily extended to binary dependent variable regression models. However, the LPM poses several limitations.

15.2.2 Limitations of the Model

The major limitations of the LPM are discussed below.

i) Non-Normality of u_i

Since the dependent variable (X_i) assumes only two values (0 and 1), the error term (u_i) also takes only two values. It implies that the error term follows the Bernoulli distribution. As a result, it is not normally distributed.

Recall from Unit 4 that the OLS method does not require the error term (u_i) to be normally distributed. However, we assumed u_i to be normally distributed for the

purpose of hypothesis testing. But the assumption of normality for u_i is not correct for LPM because, like Y_i , the error term u_i also takes only two values; that is, they also follow the Bernoulli distribution. Non-fulfillment of the normality assumption may not be so critical as it appears, because for obtaining unbiased OLS estimators, u_i need not be normally distributed. Besides, as the sample size increases indefinitely, statistical theory shows that the OLS estimators tend to be normally distributed. As a result, in large samples the statistical inference of the LPM will follow the usual OLS procedure under the normality assumption.

ii) Heteroscedasticity of u_i

Remember that for a Bernoulli distribution the mean and variance are p_i and $p_i(1 - p_i)$ respectively. Here, p_i is the probability of success, which implies that the variance is a function of the mean. Thus, the error variance is heteroscedastic. However,

$$P_i = E(Y_i | X_i, D_i = 1) = \alpha_0 + \alpha_1 X_i + \alpha_2 \quad \dots (15.5)$$

In (15.5), the mean of Y_i , that is, P_i depends on the explanatory variable X_i . Therefore, $var(u_i)$ also depends on the explanatory variables.

iii) Value of $\hat{Y}_i$ remains outside the range

We mentioned earlier that $\hat{Y}_i$ indicates the probability of occurrence, p_i . However, the OLS estimator gives no guarantee that the probability will be between 0 and 1 (See Fig. 15.1). This is because the probability increases linearly with explanatory variables.

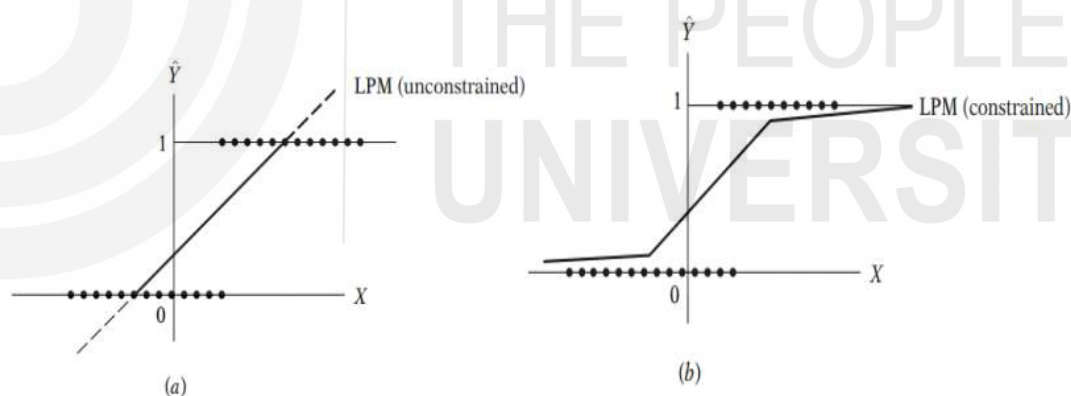


Fig. 15.1: Linear Probability Models

There are two ways to get rid of this problem: (i) Estimate the LPM by the usual OLS method and find out whether the estimated $\hat{Y}_i$ remains between 0 and 1. If for some observations $\hat{Y}_i$ is less than 0 (that is, negative), we assume them to be zero. Similarly, for some observations, if $\hat{Y}_i$ is greater than 1, we assume them to be 1. The second solution is to adopt some estimation technique such as Logit and Probit, that will guarantee that the estimated conditional probabilities $\hat{Y}_i$ remains between 0 and 1.

The scatter plot depicted in Fig. 15.1 is drawn on the basis of data given in Table 15.1. We observe that for some values of X_i , the predicted probability ($\hat{P}_i$) lies

outside the limiting values, i.e., $0 \leq \hat{P}_i \leq 1$. This violates an important axiom of probability theory that probability ranges between 0 and 1.

Table 15.1: Hypothetical Data on House Ownership Status

Household	Y	X	Household	Y	X
1	0	8	21	1	24
2	1	20	22	0	16
3	1	18	23	0	12
4	0	11	24	0	11
5	0	12	25	0	16
6	1	19	26	0	11
7	1	20	27	1	20
8	0	13	28	0	18
9	1	20	29	0	11
10	0	10	30	0	10
11	1	17	31	1	17
12	1	18	32	0	13
13	0	14	33	1	21
14	0	25	34	1	20
15	1	6	35	0	11
16	1	19	36	0	8
17	1	16	37	1	17
18	0	10	38	1	16
19	0	8	39	0	7
20	1	18	40	1	17

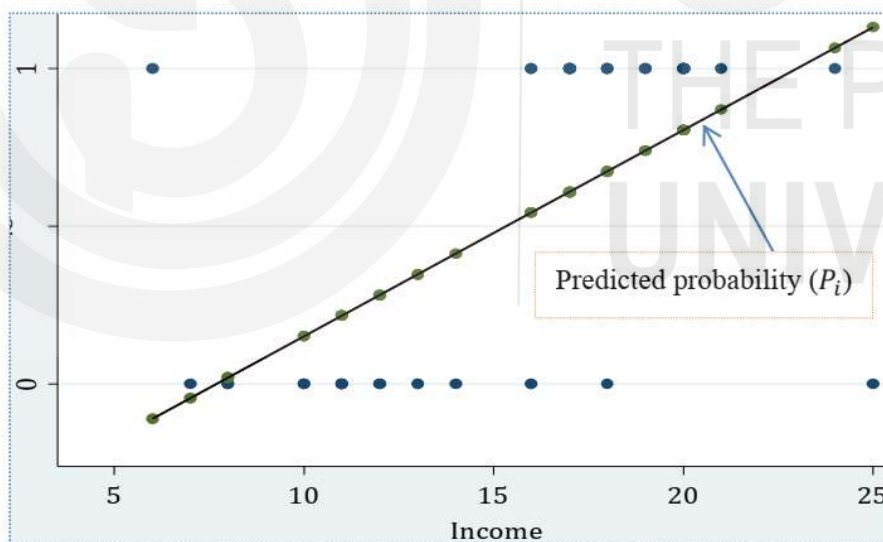


Fig. 15.2: Scatter Plot of Y_i and $\hat{Y}_i$ in LPM

iv) R^2 is not Appropriate

Corresponding to the value of explanatory variables (X), the dependent variable (Y) is either 0 or 1. Therefore, all the Y values will either lie along the X -axis or along the line corresponding to Y equals 1. Therefore, LPM is not expected to fit such a scatter plot very well. As a result, computed R^2 is of limited value in the dichotomous response models.

In Fig. 15.1, we plotted the actual values Y_i , and the predicted values $\hat{Y}_i$. Corresponding to a given X , the value of Y is either 0 or 1. Therefore, all the Y values will either lie along the X axis or along the line corresponding to 1. That is, when we regress Y_i on the given explanatory variable (X_i) in the case of linear regression model, we will get the fitted regression line given above (i.e., Fig. 15.1). Therefore, generally LPM is not expected to fit such a scatter diagram well. As a result, the conventionally computed R^2 is likely to be much lower for such models. In most practical applications, the R^2 ranges between 0.2 to 0.6.

15.2.3 Interpretation of LPM

(Example 15.1)

To illustrate the method of Linear Probability Model and the points raised above, let us assume that we want to study house ownership status (Y) of households in a given town. Assume that we have data for 40 households on house ownership status and income of households (X) (in thousands of rupees) (see Table 15.1).

Based on the data given in Table 15.1, we estimate the LPM by the OLS method. The results obtained are as follows:

$$\begin{aligned}\hat{Y} &= -0.5 + 0.0652 X_i & \dots (15.6) \\ \text{SE} &= (0.2096) \quad (0.0134) \\ t\text{-ratio} & (-)2.39 \quad 4.88 & R^2 = 0.3854\end{aligned}$$

Let us interpret the above result.

The intercept of $(-) 0.5$ gives the probability that a household with zero income will own a house. Since this value is negative, and since probability cannot be negative, we treat this value as zero. The slope value of 0.0652 means that for a unit change in income (here, Rupees 1000), the probability of owning a house increases by 0.0652 or 6.52 percent.

Table 15.2: Actual Y_i and predicted $\hat{Y}_i$

Y	$\hat{Y}_i$	Y	$\hat{Y}_i$	Y	$\hat{Y}_i$	Y	$\hat{Y}_i$
0	0.022	1	0.609	1	1.065	1	0.609
1	0.804	1	0.674	0	0.543	0	0.348
1	0.674	0	0.413	0	0.280	1	0.869
0	0.217	0	1.130	0	0.217	1	0.804
0	0.283	1	-0.109	0	0.543	0	0.217
1	0.739	1	0.739	0	0.217	0	0.022
1	0.804	1	0.543	1	0.804	1	0.609
0	0.348	0	0.152	0	0.674	1	0.543
1	0.804	0	0.022	0	0.217	0	-0.043
0	0.152	1	0.674	0	0.152	1	0.609

Table 15.2 shows the estimated probabilities, $\hat{Y}_i$, for various income levels. The most noticeable feature of Table 15.2 is that two estimated values are negative,

and two values are greater than 1. This is one reason why the LPM is not recommended.

Check Your Progress 1

- 1) A linear probability model of a hypothetical data set (sample size $n = 200$) yields the following results:

$$\hat{Y}_i = \frac{-0.084}{(0.021)} + \frac{0.0017}{(0.0004)} \text{Income}_i + \frac{0.008}{(0.0032)} D_i$$

$$t = (-4.00) \quad (4.25) \quad (2.5) \quad R^2 = 0.882$$

$Y_i = 1$ for ownership of a house, and 0 otherwise; $D_i = 1$ for male and 0 for female; income is in Rs.'000 and values in parentheses are standard errors and t-ratios.

- Interpret the model.
- Interpret the intercept term.
- Interpret the coefficient of the variable 'income'.
- Interpret the coefficient of the dummy variable, D_i .

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- 2) What are the limitations of the linear probability model?

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15.3 LOGIT MODEL

As we saw in the previous section, predicted values of the LPM model could be outside the range. Therefore, instead of depending on the LPM, we need a (probability) model that has the following two features:

- As X_i increases, $p_i = E(Y_i = 1|X)$ increases but never steps outside the $[0 - 1]$ interval.
- The relationship between p_i and X_i is non-linear. As X_i gets smaller, $\hat{Y}_i$ tends to 0 (that is, it approaches 0 at slower and slower rate). On the other hand, as X_i gets very large, $\hat{Y}_i$ tends to 1.

The logit model (logit is an abbreviation for logistic unit) is an option before us. In this model, we can limit the estimated probabilities inside the (0–1) range. In the logit model (as well as the probit model, which we will study in the next

section), we assume that there is an underlying latent variable (y^*) which is not observable, but it manifests in a dichotomous realization. For example, the latent variable is ‘the ability to get employed’, which is not observable. What we observe is whether a person is employed ($y = 1$) or not employed ($y = 0$). Thus, $y_i = 1, \text{ if } y^* > 0$ and $y_i = 0, \text{ if } y^* \leq 0$.

In the logit model, the probability of occurrence of the event is defined as

$$p_i = E(Y = 1|X_i) = \frac{1}{1+e^{-(\alpha_0+\alpha_1 X_i)}} \quad \dots (15.7)$$

If we let $Z_i = \alpha_0 + \alpha_1 X_i$, then equation (15.7) can be expressed as

$$p_i = E(Y = 1|X_i) = \frac{1}{1+e^{-Z_i}} = \frac{e^{Z_i}}{1+e^{Z_i}} \quad \dots (15.8)$$

Equation (15.8) is known as the (cumulative) logistic distribution function which satisfies the two conditions stated in the beginning of this Section. Since p_i gives the probability of owning a house, the probability of ‘not owning a house’, $(1 - p_i)$, is:

$$1 - p_i = \frac{1}{1+e^{Z_i}} \quad \dots (15.9)$$

If we divide equation (15.8) by equation (15.9), we obtain

$$\frac{p_i}{1-p_i} = \frac{e^{Z_i}/(1+e^{Z_i})}{1/(1+e^{Z_i})} = e^{Z_i} = e^{\alpha_0+\alpha_1 X_i} \quad \dots (15.10)$$

Equation (15.10) is the ‘Odds-Ratio’ in favour of owning a house. It is the ratio of the probability that ‘a family will own a house’ to the probability that ‘it will not own a house’. Thus, for example, if $p_i = 0.8$, the odds-ratio is 4. It implies that the odds are 4 to 1 (since $\frac{p_i}{1-p_i} = \frac{0.8}{1-0.8} = \frac{0.8}{0.2} = 4$) in favour of the family owning a house.

If we take the natural logarithm of equation (15.10), we obtain

$$L_i = \ln\left(\frac{p_i}{1-p_i}\right) = \ln\left(\frac{1+e^{Z_i}}{1+e^{-Z_i}}\right) = Z_i = \alpha_0 + \alpha_1 X_i \quad \dots (15.11)$$

The ‘log of the odds-ratio’ (L_i), given in equation (15.11) is linear in parameters and linear in variables.

15.3.1 Features of Logit Model

We point out the features of the logit model below.

- (i) In equation (15.11), as Z varies from $-\infty$ to $+\infty$, the ‘log of the odds-ratio’ (L_i), varies from $-\infty$ to $+\infty$. That is, although the probability remains between 0 and 1, the explanatory variable can take any value between $-\infty$ and $+\infty$.
- (ii) Although L is linear in X , the probability (p_i) is not linear. Although we have included only a single explanatory variable, in the preceding model, we can add as many explanatory variables as required by the underlying theory.

- (iii) The interpretation of the logit model given in equation (15.11) is as follows: The slope coefficient measures the change in L for a unit change in the explanatory variable. The slope coefficient tells how the log-odds in favor of owning a house change as income changes by a unit. The intercept is the value of the log-odds in favor of owning a house if income is zero. Like most interpretations of intercepts, this interpretation may not have any physical meaning.
- (iv) Remember that the odds-ratio is not the probability of the occurrence of the event (such as owning a house). Suppose we want to find out the probability of owning a house at a certain level of income, say, X^* , from equation (15.7). The question is: How do we estimate α_0 and α_1 . The answer is given below.

15.3.2 Estimation of Logit Model

Let us add a stochastic error term to equation (15.11). We obtain

$$L_i = \ln\left(\frac{p_i}{1-p_i}\right) = \alpha_0 + \alpha_1 X_i + u_i \quad \dots (15.12)$$

To estimate equation (15.12), we need data of X_i and logit (L_i). This depends on the type of data we have for analysis. There could be two types of data:

- (i) Data at the individual or micro level and
- (ii) Grouped or replicated data.

(i) Data at the individual or micro level

From equation (15.12), $p_i = 1$ if a family owns a house, and $p_i = 0$ if it does not own a house. Thus,

$$L_i = \ln\left(\frac{1}{1-1}\right) = \ln\left(\frac{1}{0}\right), \text{ if a person owns a house}$$

$$L_i = \ln\left(\frac{0}{1-0}\right) = \ln\left(\frac{0}{1}\right), \text{ if a person does not own a house}$$

These imply that if we have data at the micro or, individual level, we cannot estimate equation (15.12) by the standard OLS (ordinary least squares). In this situation, we may have to resort to **non-linear estimation procedure** using the maximum likelihood (ML) method.

(ii) Grouped or Replicated Data

Here, for a given explanatory variable, observations with equal values are grouped together. Such a model is called Grouped Logit or GLogit. For example, in the model represented by equation (15.12), people who earn the same disposable income are grouped together. Corresponding to each group of income level X_i , there are N_i families, among whom n_i are homeowners ($n_i < N_i$). Therefore,

$$\hat{p}_i = \frac{n_i}{N_i} \quad \dots (15.13)$$